

Shinde Aditya

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer specializing in GenAI Systems and Inference Optimization. Proven track record of building scalable real-time AI systems and architecting low-latency RAG pipelines. Open-source contributor (PyTorch ecosystem) with deep expertise in mathematical optimization and video processing.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, JavaScript, TypeScript, Swift
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, FastAPI, Node.js, React
- **AI & Machine Learning:** Large Language Models (LLMs), Agentic AI Systems, Retrieval-Augmented Generation (RAG), LoRA Fine-tuning, NLP, Computer Vision (YOLO, OpenCV)
- **Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, ChromaDB, Pinecone
- **Cloud & Tools:** AWS (EC2, S3), Docker, Git, Hugging Face Hub, OpenAI API
- **Other Competencies:** Distributed Systems, Microservices, System Design, API Integration, Unit Testing, Monitoring

EXPERIENCE

- **AI Engineer** Aug 2025 – Present
TechOptima.ai Hyderabad, India
 - **Real-Time Multimodal Systems:** Architected an interactive AI Avatar system integrating LLMs with Speech-to-Video (S2V) synthesis, achieving low-latency audio-visual synchronization for seamless educational content delivery.
 - **Crowd Density & Safety Analytics:** Engineered a real-time computer vision pipeline for Railway Platform Safety, utilizing YOLO-based object detection to quantify passenger density and identify available platform space.
 - **System Optimization:** Optimized model serving performance by implementing asynchronous request handling in FastAPI and containerizing the inference engine using Docker for scalable deployment on AWS EC2.
 - **Scalable API Architecture:** Designed an asynchronous FastAPI serving layer capable of handling high-volume concurrent requests, ensuring responsive interactions for web and mobile clients.
 - **MLOps & CI/CD:** Automated model deployment pipelines using GitHub Actions, significantly reducing rollout time and ensuring system reliability via Weights & Biases monitoring.
- **Machine Learning Engineer Intern** Feb 2024 – June 2024
Conquest Technology Solutions Hyderabad, India
 - **Automated Data Pipelines:** Engineered automated data extraction and transformation scripts using Python and SQL, replacing manual workflows and reducing reporting latency by 45%.
 - **Predictive Modeling:** Built and validated baseline ML models (Classification/Regression) using Scikit-learn, performing rigorous EDA on large-scale transactional datasets to optimize feature selection.
 - **Data Engineering:** Collaborated with senior engineers to optimize SQL queries and validate data integrity, ensuring high-quality inputs for downstream machine learning tasks.

PROJECTS

- **torch-secorder (Deep Learning Optimization)** GitHub — Docs — PyPI
Python, PyTorch, Sequence Modeling, Poetry, CI/CD
 - **Core Research:** Developed a PyTorch-native library for Second-Order Optimization, implementing Hessian-Vector Products (HVP) and Gauss-Newton matrix computations for curvature analysis.
 - **Performance:** Implemented Hutchinson's method for efficient Trace Estimation, enabling advanced convergence analysis on consumer-grade GPUs.
 - **Adoption:** Open-sourced the library with distributed computation support and automatic parameter handling, targeting researchers in model optimization.

- **DevFlow-AI** GitHub — Docs — Marketplace
Python, FastAPI, LangChain, TypeScript, ChromaDB, Tree-sitter, Docker, OpenAI
 - **Architecture:** Engineered a privacy-centric Agentic RAG system that runs locally, utilizing Tree-sitter for robust AST parsing and GitPython for evolution analysis.
 - **Tech Stack:** Built a FastAPI backend with ChromaDB for vector storage, integrated directly into VS Code via a TypeScript frontend.
 - **Innovation:** Designed a "Ask My Codebase" feature that correlates code chunks with internal documentation and commit history to explain legacy system architecture.
- **Coral (Video Object Tracking & Segmentation)** GitHub
Python, OpenCV, PyTorch, YOLOv8, U-Net
 - **Computer Vision Pipeline:** Built an end-to-end Video Segmentation & Tracking system utilizing YOLOv8/v9 for detection and U-Net for pixel-level segmentation.
 - **Impact:** Engineered automated area quantification tools to analyze biological footage, a workflow directly applicable to AI-based Video Editing and object removal tasks.

EDUCATION

- **Bachelor of Technology in Data Science & Artificial Intelligence** Sep 2021 – Oct 2025
Woxsen University Hyderabad, India

ACHIEVEMENTS

- **Published Research – AI for Coral Species Classification:** Co-authored a peer-reviewed paper titled “*AI Infused Underwater Analysis for Enhanced Coral Species Classification*” IndiaCom 2025, IEEE. 
- **Best Presentation Award – OIST Hackathon:** Led a cross-functional team to develop a real-time language translation tool using speech-to-text APIs and transformer-based models; recognized for clarity, innovation and technical depth.
- **PyPI Author – Open-Source ML Tools:** Published multiple production-grade Python packages to PyPI (`ml-assert`, `torch-secorder`) with CI/CD pipelines, documentation, and adoption-ready CLI tools.